

# INNOTRACK INTERNSHIP PROGRAM - 2026



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- **Project Name – REAL TIME HEART RATE ANOMALY DETECTOR**
- **Theme – MACHINE LEARNING IN HEALTHCARE**
- **VTU ID – 27976**



# REAL TIME HEART RATE ANOMALY DETECTOR



## INTRODUCTION :

*Real-time heart rate anomaly detection is an important application of Machine Learning in the healthcare sector. The HeartSync system predicts and monitors a patient's heart condition using factors such as heart rate, SpO<sub>2</sub>, blood pressure, activity type, and stress level.*

*The system helps doctors and caretakers analyze patient vital trends and make informed medical decisions. By utilizing data-driven detection techniques, the project contributes to improved healthcare monitoring, early diagnosis, and patient safety.*

# Idea and solution:

## IDEA

*The idea behind this project is to build an intelligent system capable of detecting anomalies in real-time heart rate data using Machine Learning algorithms. By analyzing patient vitals and behavioral factors, the system can identify abnormal patterns and provide valuable health insights.*

## SOLUTION

*The proposed solution is a web-based Real-Time Heart Rate Anomaly Detector developed using Python, Scikit-learn, SQLite, and Flask. Users can enter patient-related information through a simple interface, and the system detects anomalies and predicts stress levels using trained machine learning models*

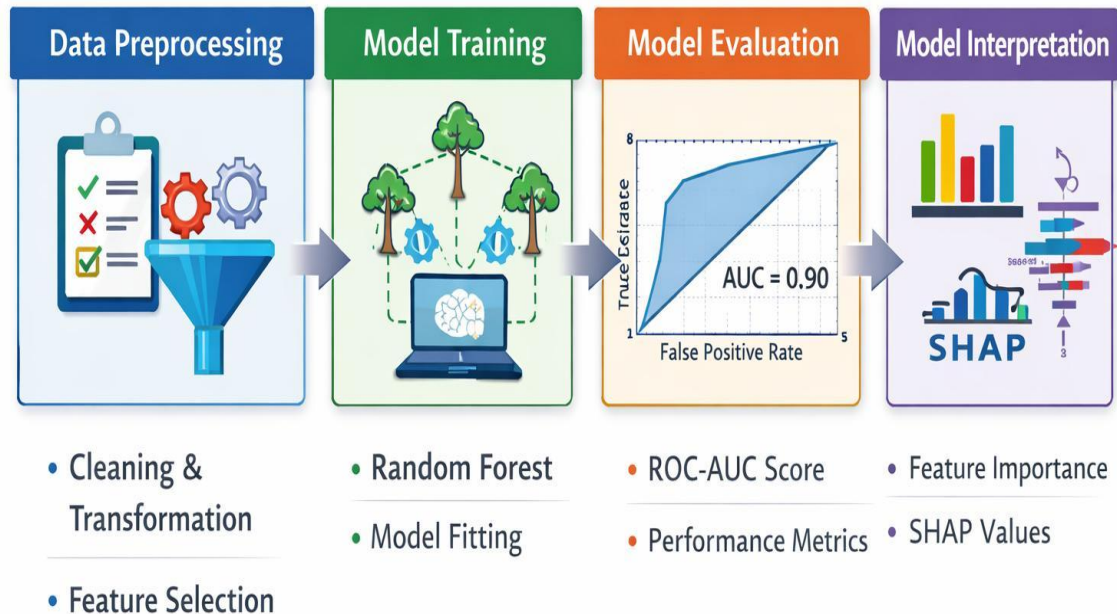
# Uniqueness:

- *Detects heart rate anomalies using Machine Learning algorithms in real time.*
- *Provides quick and automated anomaly detection based on multiple vital parameters.*
- *Combines data analysis, machine learning, database management, and web development in a single application.*
- *Uses historical patient vital data to generate meaningful health insights.*
- *User-friendly web interface that allows easy interaction without technical knowledge.*
- *Helps identify patients who may require immediate medical attention at an early stage.*

# TECHNICAL APPROACH



## Machine Learning Pipeline for Healthcare Data



## Technologies to be Used:

**Python** was used as the core programming language to develop the machine learning model and handle data processing tasks. **Pandas, NumPy, and Scikit-learn** were used for data analysis, preprocessing, model training, and prediction. **Flask, HTML, CSS, and SQLite** were used to build the web application, manage patient data, and provide real-time anomaly detection through a user-friendly interface

## Methodology:

Dataset Collection, Data Cleaning and Preprocessing, Exploratory Data Analysis (EDA), Feature Selection, Model Training, Model Evaluation, Model Saving (.pkl), SQLite Database Integration, Flask Web Application Deployment

# FEASIBILITY AND VIABILITY



## **Technical Feasibility**

The project is technically feasible because it uses proven machine learning algorithms and widely adopted open-source technologies. The implementation can be easily maintained and extended.

## **Economic Feasibility**

The project has minimal development cost because all software tools and libraries used are open source. No expensive infrastructure or licensing costs are required.

## **Operational Feasibility**

The application is simple to use and requires minimal technical knowledge. Doctors and caretakers can interact with the system through an intuitive web interface.

## **Viability**

- Provides real-world value in healthcare.
- Helps identify at-risk patients early.
- Supports data-driven medical decisions.
- Easy to use and maintain.
- Can be updated with new patient data.
- Improves patient health monitoring.
- Compatible with future healthcare systems.
- Sustainable and scalable solution.

# CHALLENGES FACED



*During the development of the Real-Time Heart Rate Anomaly Detector, several challenges were encountered.*

*One of the primary challenges was understanding and preprocessing the dataset. The data contained multiple vital features that required cleaning, transformation, and proper encoding before being used for machine learning.*

*Another challenge was selecting the most suitable machine learning algorithm. Different models were evaluated and compared to identify the model that produced the most accurate anomaly detection results. Hyperparameter tuning was also performed to improve performance and reduce prediction errors.*

*Integration of the trained model into a Flask web application presented additional challenges. Ensuring smooth interaction between the user interface, SQLite database, and machine learning backend required extensive testing and debugging. Dependency management, environment configuration, and deployment-related issues were also addressed throughout the project lifecycle.*

# FUTURE ENHANCEMENTS



*The Real-Time Heart Rate Anomaly Detector can be further enhanced by incorporating advanced machine learning algorithms such as LSTM, XGBoost, and Neural Networks to improve anomaly detection accuracy.*

*Additional patient-related vital features can be included to make the model more comprehensive.*

***Future versions of the project may include:***

- *Interactive dashboards for health analytics.*
- *Personalized health recommendations.*
- *Cloud deployment for improved accessibility.*
- *Mobile application integration.*
- *Real-time IoT sensor connectivity.*
- *Automated report generation.*
- *Hospital Management System integration.*
- *Doctor and caretaker notification system.*



# IMPACT AND BENEFITS

## ***Impacts :***

- *Enables early identification of patients who may require immediate medical attention.*
- *Promotes data-driven decision-making in healthcare institutions.*
- *Improves monitoring and analysis of patient vital trends.*
- *Demonstrates the practical application of Machine Learning in healthcare.*
- *Contributes to better medical planning and improved patient outcomes*

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## ***Benefits :***

- *Provides quick and accurate detection of heart rate anomalies.*
- *Helps doctors offer personalized guidance and treatment support.*
- *Assists caretakers in understanding and monitoring patient health.*
- *Reduces manual effort involved in continuous patient monitoring.*
- *Offers a user-friendly web interface for easy accessibility and usage.*



# CONCLUSION

- Successfully developed a Machine Learning-based Real-Time Heart Rate Anomaly Detector.
- Applied data preprocessing, model training, evaluation, and deployment techniques.
- Achieved reliable anomaly detection performance using machine learning algorithms.
- Integrated the trained model into a Flask web application with SQLite database support.
- Demonstrated the potential of Artificial Intelligence in improving healthcare decision-making.
- Successfully transformed patient vital data into meaningful health predictions.
- Demonstrated the complete machine learning project lifecycle.
- Showcased practical implementation of anomaly detection analytics.
- Built a scalable foundation for future healthcare applications.
- Established a reliable and user-friendly heart rate anomaly detection platform.

**THANK YOU**